

# Cheng Wu

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## EDUCATION

### Columbia University, Engineering

Master of Science in Data Science, GPA: 3.80/4.00

New York, NY  
Expected Dec 2026

### University of Illinois at Urbana-Champaign

Bachelor of Science in Econometrics & Quantitative Economics, Minor in Statistics, GPA: 3.86/4.00

Champaign, IL  
May 2025

Selected Coursework: Market Microstructure & Algorithmic Trading, Optimization & Operations Research, Monte Carlo Methods, Statistical Learning

## WORK EXPERIENCE

### Compliance Risk Analyst Intern, Tarte Cosmetics | New York, NY

Oct 2025 - May 2026

- Built a **Python/SQL compliance risk-scoring model** across 7 legacy systems, replacing manual case-by-case triage with **risk-prioritized exception review queues**
- Calibrated **escalation thresholds** using historical reviewer overrides and Precision@K, Recall, and Lift, reducing low-yield alerts by **25%** while preserving all material exceptions in the highest-priority review tier.
- Identified 12 recurring workflow and control breakdowns responsible for **40% of audit delays** through Pareto root-cause analysis, directing remediation toward the highest-frequency issues
- Developed **Power BI risk-monitoring dashboards** for exception volume, review aging, and recurring vendor patterns, contributing to a **22% reduction in repeat compliance issues**

### Data Scientist Intern, Donglai Natural BioTech | Champaign, IL

Jun 2024 - Sep 2024

- Integrated customer, transaction, and digital-channel data across **10K+ customers** and developed RFM segmentation and repeat-purchase **propensity models**, achieving **0.78 ROC-AUC**
- Constructed a weekly **Marketing Mix Model** across 5+ channels using adstock, saturation, and regularized regression; translated model output into MROI and budget-allocation scenarios with **~10% higher modeled incremental contribution** under fixed-budget constraints
- Estimated price and promotion elasticities across **50+ SKUs** and designed **3+ A/B tests** using power/MDE and guardrail metrics, supporting customer-facing changes that delivered a **mid-single-digit relative conversion lift**
- Automated invoice/vendor review using LangChain extraction and **dbt/SQL** data-quality controls, reducing daily manual review time by 30%

### Data Analyst Intern, Zheshang Securities | Hangzhou, China

May 2023 - Sep 2023

- Standardized **SQL risk-reporting and reconciliation logic** across 15+ valuation, exposure, and risk fields for **50K+ financial records**, reducing manual reconciliation effort by **70%** through exception-based review
- Applied historical-baseline thresholds and **z-score anomaly detection** to distinguish data-quality issues from material position changes requiring analyst escalation
- Performed **scenario and sensitivity analysis** for 10+ quarterly investment reviews, translating changes in portfolio assumptions into impact estimates and stakeholder recommendations

## SOCIAL IMPACT EXPERIENCE

### Lead Data Scientist, Innovative Bloom Foundation | Shanghai, China

May 2019 - Jun 2024

- Architected a 6-year **SQL/R longitudinal database** covering **1,200+ participants** and developed risk-tier segmentation to prioritize limited program resources
- Leveraged **Propensity Score Matching, Difference-in-Differences, and heterogeneous treatment-effect analysis** to identify high-response groups and guide intervention allocation, contributing to a **~19% reduction in high-risk participant attrition**

## PROJECTS

### [Applied AI Project - Multi-Document Intelligence](#)

Aug 2025 - Jan 2026

- Engineered and evaluated a multi-document **RAG** system using Sentence-Transformers, FAISS, and LangChain; improved grounded-answer accuracy from **62% to 87%** across a structured 27-case evaluation benchmark

### [Open-Data Econometrics - San Francisco Crime](#)

Sep 2024 - Jan 2025

- Linked **913K+ crime records** with Census/ACS data using **PySpark spatial joins** and benchmarked fixed-effects, Poisson/Negative Binomial, and ARIMAX/SARIMAX models across socioeconomic and temporal specifications

## SKILLS

Programming & DBs: Python, SQL (PostgreSQL/MySQL), R, STATA, PySpark, dbt, MongoDB, Git, Excel/VBA ML, AI & Frameworks: scikit-learn, PyTorch, LangChain, FAISS, Sentence-Transformers, pandas, NumPy, SciPy

Modeling & Methodologies: A/B Testing, Causal Inference (DiD/PSM), Time Series (ARIMA), Marketing Mix Modeling (MMM), Log-Log Elasticity

Analytics & Visualization: Power BI, Tableau, Streamlit, Anomaly Detection, Scenario & Sensitivity Stress Analysis